A LIVING CLIMATE SIMULATION FOR EVERY GENERATION

EARTH ENGINE

“Don't read climate change. Live it.”

A living simulation of a West African settlement — evolving in real time, over decades. Full project report prepared for the Climate Change Hackathon, 2026.

CLIMATE CHANGE HACKATHON · 2026

Contents

A full walkthrough of the world, the problem, and the platform.

01 · THE WORLD

Ama is 12. Her life works.

Her parents farm the same land her grandparents farmed. The harvest has always been enough. Every morning, she walks this path to school.

Ama's village sits along a river that has never, in living memory, run dry. The rains arrive in the same weeks every year — early enough to plant, heavy enough to fill the reservoir, gone in time for harvest. Her father grows maize and cassava on the same plot his father worked. Her mother sells vegetables at the market by the bridge. None of this required luck. It required only that the climate behave the way it always had.

This stability is what makes Ama's daily routine possible: an hour's walk past the school sign that reads "Education Is Our Hope For A Better Tomorrow," full classrooms, a village economy that runs on predictable seasons.

● Farming community

Multi-generational subsistence and cash-crop farming along a stable river system.

● Reliable rainfall

Predictable wet and dry seasons that the whole local economy is built around.

● Full classrooms

Children attend school because their labor isn't yet needed to keep the household fed.

This baseline — stable rainfall, stable harvest, stable schooling — is the starting condition Earth Engine simulates before any climate shock is introduced.

02 · THE PROBLEM

Then, one season, everything changed.

Climate disruption rarely arrives as a single dramatic event. It arrives as three quiet failures, each one making the next one worse.

The rain changed

Clouds drifted past without releasing water. The season that should have brought rain stayed dry. Planting windows that farmers had used for generations no longer lined up with when the rain actually came.

The river dried

The water that fed the whole village shrank to a thin trickle across cracked, exposed earth. Wells that had never failed began to run low by mid-season.

The harvest failed

For the first time in memory, the maize came in brittle and stunted. Not enough to sell. Barely enough to eat. A single bad season converted a self-sufficient household into one that needed help.

What makes this pattern dangerous is how interconnected it is. A rainfall shift is a meteorological fact. A dried river is a hydrological consequence. A failed harvest is an economic one. None of these systems are separate in real life — but most educational tools about climate change teach them as separate facts, in separate chapters, disconnected from one another. Earth Engine's core premise is that this interconnection is the actual lesson.

03 · THE HUMAN COST

Families began to leave.
Ama stopped going to school.

With less water and smaller harvests, food prices rose faster than families could adapt. Her family needed help on what was left of the farm.

Climate shocks don't affect every household in a village the same way. They ripple outward differently depending on who you are, what you own, and how much of a buffer you had to begin with. To make that concrete, consider three residents of the same village, in the same season, living through the same drought.

A

Ama, age 12 — Child

HOUSEHOLD: FARMING FAMILY

Her family needed help on what was left of the farm. Chores that used to happen after school now happen instead of school. She is not out of the household's plans for the future — but she is out of the classroom for as long as the harvest stays this thin.

K

Kwame, age 45 — Farmer

HOUSEHOLD: AMA'S FATHER

A lifetime of knowledge about when to plant and when to expect rain stopped matching reality. With yields down and no irrigation to fall back on, he faces a choice made by so many heads of household in a drought: stay and gamble on next season, or leave to find wage work elsewhere and send money home.

N

Nana, age 68 — Elder

HOUSEHOLD: EXTENDED FAMILY, FIXED INCOME

Elders are often the least mobile and most dependent on local food prices staying stable. As maize prices climb with scarcity, a fixed income buys less every week. Nana cannot easily leave, cannot easily farm, and absorbs the shock differently than a working-age adult or a child.

> *This is not a hypothetical child.*
> *This is what happens to millions, every year.*

This is climate change.

Not a headline. A childhood.

What if anyone could predict these consequences before they happen?

04 · INTRODUCING

EARTH ENGINE

A living simulation of a West African settlement — evolving in real time, over decades.

Don't read climate change. Live it.

Earth Engine takes the chain of cause and effect you just read — rainfall, river, harvest, household — and turns it into something a player can act on, not just read about. Instead of narrating what happened to Ama's village, it lets a player govern a village like it, watch the same forces play out, and try to change the outcome.

05 · CORE IDEA

General overview — the core idea.

Earth Engine is an interactive climate simulation platform that transforms climate education from passive learning into immersive experience. Rather than teaching climate change through textbooks, presentations, or statistics, the platform allows users to step into a living virtual community where they can observe, experiment, and experience the long-term consequences of their decisions.

Set within a realistic West African settlement, every aspect of the community — from its people and economy to its forests, rivers, farms, schools, and infrastructure — is interconnected. As time progresses, climate events such as droughts, floods, rising temperatures, and environmental degradation continuously reshape the community, requiring users to make informed decisions that balance economic growth, environmental sustainability, and human well-being.

Learning by consequence, not by fact

At the heart of Earth Engine is the belief that people learn best by experiencing consequences rather than memorizing facts. Users are not simply presented with information; they become decision-makers responsible for guiding a community through decades of environmental change. Every policy implemented — whether investing in renewable energy, protecting forests, improving water management, or expanding agriculture — produces visible social, economic, and environmental outcomes that unfold over simulated years. This cause-and-effect approach enables users of all ages, from children to policymakers, to develop an intuitive understanding of climate change and resilience.

Where this leads

By combining real-time simulation, intelligent virtual citizens, and interactive decision-making within a game-inspired environment, Earth Engine bridges the gap between education, technology, and climate awareness. Beyond serving as an engaging educational tool, the platform establishes a foundation for future applications in schools, public awareness campaigns, policy evaluation, community planning, and digital twin simulations. Its long-term vision is to make complex climate systems understandable, interactive, and accessible to everyone, empowering individuals and organizations to make more informed decisions about the future of our planet.

> *Earth Engine is not a game about climate change; it is a* **living simulation where climate change becomes an experience** *— allowing anyone, from a 10-year-old student to a government policymaker, to understand the consequences of today's decisions by witnessing tomorrow unfold before their eyes.*

06 · THE SOLUTION

One world. Every system connected.

The settlement is rendered as a single connected map spanning agriculture, water, forests, energy, education, health, economy, people, and infrastructure. None of these are independent mini-games or isolated modules — they read from and write to the same underlying world state, so a change in one shows up as a consequence in another.

Core design principles

1 **Citizens make independent decisions**

Simulated residents don't wait for player instructions — they act on their own incentives (income, hunger, safety), the same way real households do.

2 **Weather changes; economies grow or shrink**

Rainfall and temperature patterns drive agricultural yield, which drives household income, which drives the whole local economy up or down.

3 **Forests disappear or regrow; floods happen**

Land-use choices feed back into the water cycle — deforestation increases flood and drought risk in later years of the same run.

4 **Players learn by observing consequence — never a lecture**

There is no dialogue box explaining climate science. The lesson is what the player watches unfold after their own decision.

07 · THE EXPERIENCE

You don't control people.
You shape the conditions they live in.

Earth Engine deliberately withholds direct control over individual citizens. A player cannot order Ama back to school or force Kwame to stay on the farm. What a player *can* do is change the conditions those decisions get made inside — the same lever real policy actually pulls.

Interventions fall into two families: those that build long-term resilience, and those that offer short-term relief at a longer-term cost. Both are available. Both are used by real communities and real governments. Earth Engine doesn't hide the second category — it lets players discover why it's tempting and what it costs.

Resilience-building actions

- Plant forests
- Build solar
- Irrigation
- Recycle
- Education
- Disaster prep

Short-term / high-cost actions

- Burn waste
- Fossil fuels

Both categories move the simulation forward — but on very different timelines. Fossil fuel use might stabilize a village's energy needs for a season while quietly degrading air quality and long-term climate stability. That trade-off is the point: it's the same trade-off real communities and policymakers face.

08 · THE EXPERIENCE

Every citizen is simulated. Individually.

Behind every policy are individually modeled citizens — each with an age, occupation, income, health, education, and family. A farmer reacts differently than a teacher. A child responds differently than an elder. This is what separates Earth Engine from a top-down statistics dashboard: the numbers on screen are aggregated from individual, simulated households, not scripted to move in a pre-set curve.

CITIZEN TYPE	ATTRIBUTES TRACKED
Child	Health · Income · Family
Farmer	Health · Income · Family
Teacher	Health · Income · Family
Elder	Health · Income · Family
Vendor	Health · Income · Family

Because Ama, Kwame, and Nana in the opening story map directly onto the Child, Farmer, and Elder citizen types in the simulation, a player who has just read their story can immediately recognize the same pressures acting on the simulated population in front of them — and see, over a run of simulated years, whether their policy choices change that family's trajectory.

09 · THE TECHNOLOGY

Years pass in seconds. Watch what happens.

A twelve-year time-lapse — sunrise, growth, rainfall, market activity, then flood rise, forest loss, and recovery through solar and regrowth — compressed into a continuous run: **Year 1 → Year 12.**

Growth | Rainfall / Flood | Drought stress

Live demo

https://earth-engine.netlify.app

Not fifteen apps. One living world.

Every system below shares the same simulation state and continuously affects every other system, rather than existing as a standalone screen or mini-app:

Forests	Energy
Circular Economy	Migration
Water	Agriculture
Disaster Response	Education

Brief technical overview

At a high level, Earth Engine is organized into three layers that stay deliberately simple to describe, even though the simulation running underneath is not:

Frontend

A game-like interactive environment players explore in real time, where the visible village reflects the current simulation state.

Backend

A real-time simulation engine running continuous world-state calculations — advancing weather, resources, and citizen conditions tick by tick.

AI Layer

Behaviour simulation and environmental modelling — how individual citizens act on their own incentives, and how ecosystems (forest, water, soil) respond to those actions over time.

Player sets policy → Environment updates → Citizens react → Economy shifts → Player observes outcome

This loop repeats continuously as simulated years advance, which is what lets twelve years of consequence play out in a matter of seconds on screen.

10 · WHY IT WORKS

Traditional education fades. Experience doesn't.

Climate literacy is usually delivered as information: a chart, a headline, a paragraph in a textbook. Information is easy to forget once the test is over, because there was never a personal stake in it. Earth Engine's pedagogical bet is simple — a player who watches their own decision cause Ama's family to leave the village remembers that in a way no paragraph achieves.

TRADITIONAL

1. Read
2. Watch
3. Memorize
4. Forget

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1. Interact
2. Experiment
3. Observe
4. Understand
5. **Remember**

The difference between the two columns isn't tone — it's ownership. In the traditional path, the consequence belongs to someone else, described secondhand. In Earth Engine, the consequence belongs to the player's own choice, observed firsthand inside a system they were just given the controls to.

11 · THE IMPACT

Built for classrooms. Ready for governments.

Because the underlying simulation doesn't change — only who's looking at it and what they need from it does — Earth Engine is designed to scale from a single classroom to something closer to policy infrastructure, without a rebuild at each step.

WHO IT'S FOR

- Students
- Schools
- Governments
- NGOs
- Researchers
- Climate Orgs
- Policy Makers
- Communities

PLATFORM PATHWAYS

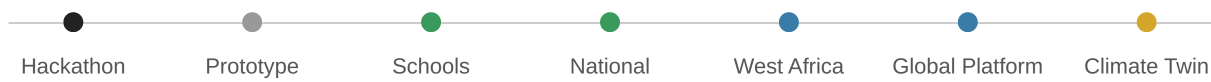
- Educational licensing
- Government planning tools
- NGO training modules
- Public awareness campaigns
- Climate preparedness sims
- Full digital twins

PATHWAY	WHAT IT LOOKS LIKE IN PRACTICE
Educational licensing	Classroom-ready builds for schools, used alongside a standard climate curriculum.
Government planning tools	Scenario testing for local officials evaluating irrigation, energy, or resettlement policy.
NGO training modules	Field-staff training on how interventions interact before they're deployed on the ground.
Public awareness campaigns	Public-facing, simplified builds designed for open exploration rather than instruction.
Climate preparedness sims	Stress-testing disaster response plans against simulated droughts and floods.
Full digital twins	Long-horizon, high-fidelity models of real settlements for ongoing planning use.

12 · THE FUTURE

From this room to a global climate twin.

The version presented at this hackathon is intentionally scoped: one settlement, one region's climate patterns, a fixed set of citizen types. Everything about the underlying design is meant to generalize past that scope.



STAGE	WHAT IT REPRESENTS
Hackathon	The current build — one settlement, core systems connected, live demo available.
Prototype	Hardened build with a stable citizen-behaviour model and refined UI.
Schools	Pilot classroom deployments paired with teacher-facing guidance.
National	Multiple regional settlement templates within a single country's climate profile.
West Africa	Expansion across the region's diverse climate zones and livelihoods.
Global Platform	Settlement templates generalized to other climates and geographies worldwide.
Climate Twin	High-fidelity, continuously updated digital twins used for real planning decisions.

We cannot change yesterday.

But together, we can simulate tomorrow.

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